



AFONSO CABOZ

Systems Architect: AI & Full-Stack Integration

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PHILOSOPHY

*"Code is a liability.
A solution is an asset."*

My guiding principle is to write the least amount of code necessary to create the most value.

TECH STACK

AI / LLM

RAG Pipelines

CrewAI

Langfuse

LiteLLM

Observability

Backend & Systems

Python

FastAPI

Django

Rust

Axum

Tauri

PostgreSQL

Frontend

Next.js

Vite

TypeScript

Tailwind CSS

Infrastructure

Docker

Terraform

CI/CD

EDUCATION

Web development technical degree

ETIC Algarve

2025 • Final grade: 18/20

An incubator for my evolution into a Solutions Architect. I applied engineering rigor to institutional challenges, culminating in the independent architecture and deployment of a resource management system still in active use. Recognized by the School Director as a professional-grade asset rather than a student assignment, this project transformed the institution's infrastructure, bridging the gap between academic study and production-ready utility.

I engineer scalable, observable, and cost-optimized production environments for AI-driven applications.

I make sense of technical complexity through **associative memory and visual thinking**. I connect new problems with familiar code patterns, past debugging experience, and technical documentation, then mentally map system components, dependencies, and data flows. This helps me identify **potential failure points and compare architectural approaches** before implementation, bringing clearer proposals to technical discussions and sharper questions to validation.

Methodology: A balance between **Deep Research/UX Mapping** and **AI-Augmented Execution** to guarantee architectural integrity with high-velocity deployment.

AI-Driven Development: I structure AI-assisted delivery through project-specific agent instructions and Serena MCP for codebase understanding and durable documentation. I select Codex for daily development and Claude Code for heavier requirements, then use pre-commit checks and CodeRabbit review as quality gates before changes move forward.

Objective: Transforming complex technical requirements into resilient, client-ready architectures.

Spiritual growth is an important part of my personal development. I practise **Reiki** and draw on **Ayurvedic principles** to guide my dietary choices.

Time in nature brings balance to everyday city life, with **sea swimming** as my preferred form of exercise. I also practise **martial arts**, making space for both physical discipline and reflection.

Selected Projects

CLIENT PROJECTS

ETIC_Algarve Resource Hub: Unified Institutional Management System

My role. Solutions Architect & Lead Engineer

Engineered a high-performance ecosystem to unify fragmented institutional resources. The architecture features advanced search with intelligent autocomplete and secure identity management via **Supabase/Google OAuth (RBAC)**. To ensure operational integrity, I implemented automated communication via **Resend API** and validated system stability through **k6 load testing** (50+ concurrent users). Deployment is fully automated using **Terraform, Docker Compose**, and a **Makefile-based CI/CD** pipeline with **Nginx SSL** termination.

SKILLS AND DELIVERABLES

CI/CD

PERFORMANCE TESTING

LOAD TESTING

TERRAFORM

DOCKER

FleetFlow: AI-Powered Autonomous Fleet Management Platform

My role. AI Solutions Architect: LLMOps & Agentic Systems

Engineered the transition of FleetFlow from simple task-driven assistants to fully autonomous, **RAG-powered agents**. Implemented a multi-layer observability architecture using **Langfuse** and **LiteLLM**, utilizing an "**LLM-as-a-Judge**" framework to reduce API costs while increasing output precision. Developed a predictive "**Fuel Intelligence Layer**" for anomaly and fraud detection via advanced algorithmic modeling. Additionally, engineered an **AI Automated Testing** pipeline to mitigate model hallucinations, ensuring a self-correcting, reliable production environment.

SKILLS AND DELIVERABLES

MLOPS

LANGFUSE

AI AGENT DEVELOPMENT

PREDICTIVE ANALYTICS

AI IMPLEMENTATION

SOFT SKILLS

Forged in high-pressure hospitality environments and refined through complex technical work, these strengths shape how I collaborate, decide, and support teams.

Empathetic Thinking

User-centric & collaborative

Out-of-the-box Perspective

Creative problem solving

Mental Visualization

Conceptual design

Associative Memory

Pattern recognition & synthesis

Present Attitude

High-pressure adaptability

Nurturing Environments

Supportive collaboration

PERSONAL PROJECTS

CyberCompass: Agentic Cyber-Ethics Learning Platform

My role. Full-Stack AI Solutions Architect

Engineered a bilingual cyber-ethics platform delivering adaptive training across disinformation, deepfake detection, cyberbullying, and online identity safety. Architected a three-service system with a **Next.js 15/React 19** frontend and dedicated **Rust/Axum** database and AI orchestration services. Implemented real-time agentic simulations over **WebSockets** with pluggable LLM providers, structured prompting, safety classification, deterministic scoring and fallbacks, plus multi-layer **Langfuse/OpenTelemetry** observability for quality, cost, and risk attribution.

SKILLS AND DELIVERABLES

RUST

AGENTIC AI & LLM ORCHESTRATION

PROMPT ENGINEERING & AI SAFETY

WEBSOCKETS & REAL-TIME

LANGFUSE OBSERVABILITY

IN Sintonia: Agentic Ayurvedic Nutrition Platform

My role. Full-Stack AI Solutions Architect

Engineered an agentic real-time platform to align dietary interventions with Ayurvedic principles. Implemented a highly efficient **Mixture-of-Experts (MoE)** architecture using hierarchical prompts and a “**root-level**” verification layer to eliminate semantic drift. Architected a high-performance **Rust** backend leveraging **ZeroClaw** for AI orchestration, utilizing **Axum** and **Utopia** for high-throughput APIs. Integrated **WebSockets** to facilitate seamless, real-time nutritional guidance and continuous data streams via a modern **Next.js/Tailwind** frontend.

SKILLS AND DELIVERABLES

AI MODEL INTEGRATION

AI AGENT DEVELOPMENT

RUST

LANGCHAIN

AI COMPLIANCE

Employment History

AI Solutions Architect | LLMOps & Agentic Systems

Feb 2026 – May 2026

VivaDrive

Project Overview: Scaling the FleetFlow AI-powered fleet management platform from reactive assistants to fully autonomous, RAG-driven agents.

Key Interventions:

- **Unified Observability & Cost Optimization:** Engineered a multi-layer observability architecture using Langfuse and LiteLLM. Implemented an “LLM-as-a-Judge” framework, reducing API costs via gpt-4o-mini while increasing output precision through total request traceability.
- **Predictive Intelligence Development:** Developed the core “Fuel Intelligence Layer,” transitioning architecture from static data processing to a predictive Data Warehouse model for automated anomaly and fraud detection.
- **Autonomous Reliability:** Engineered an AI Automated Testing pipeline to mitigate model hallucinations, creating a self-correcting ecosystem where specialized models validate autonomous agent outputs.

Result: Transformed FleetFlow into a resilient, predictive architecture capable of managing complex logistics with minimal human intervention.

Solutions Architect | Systemic Debt Recovery & MVP Deployment

Aug 2025 – Sep 2025

ETIC_Algarve

Project Overview: Rehabilitating a legacy resource management system at ETIC Algarve burdened by mounting technical debt and architectural entropy.

Key Interventions:

- **Architectural Audit:** Conducted deep-dive audits to identify root causes of systemic failure, addressing fundamental structural decay rather than superficial UI fixes.
- **Strategic Rebuild Advocacy:** Engineered a framework to pivot from high-maintenance refactoring toward a modern, scalable architecture for long-term maintainability.
- **Rapid Prototyping (48-Hour Execution):** Developed a functional, high-fidelity prototype within 48 hours to validate technical viability and secure stakeholder buy-in.
- **Stakeholder Synchronization:** Aligned complex engineering milestones with organizational objectives and institutional feedback loops.

Result: Delivered a production-ready MVP, transforming a failing legacy asset into a scalable, high-performance platform.